

Improving Customer Support Capability at Abbott Diabetes Care Using Cluster Analysis

As Abbott Diabetes Care (ADC) prepares to migrate the US consumer website into a new design system, a special challenge is presented by the volume of FAQs alone. This presents a strategic opportunity to improve customer experience and internal efficiency through unsupervised machine learning—specifically, cluster analysis. By grouping related support content, ADC can reduce redundancy, identify emerging themes, streamline regulatory review cycles while improving customer satisfaction.

Analytical Approach: Unsupervised Machine Learning

Cluster analysis is an unsupervised learning method that identifies patterns in unlabeled data. Unlike supervised learning, which predicts outcomes based on labeled examples, unsupervised learning explores structure within the data itself (Chan, 2025). In ADC’s case, the goal is not to predict a specific outcome, but to organize support content into meaningful, actionable clusters.

Unsupervised learning is ideal for “wide data” environments—those with many variables but no clear dependent outcome (Yeomans, 2015). ADC’s support ecosystem includes: FAQ web content, chat transcripts, customer support call logs, and organic search queries. These inputs are rich in language but lack labeled outcomes, making them well-suited for exploratory clustering. Recent industry applications confirm this approach: companies are using clustering to group support tickets, identify intent patterns, and streamline knowledge bases (DevRev, 2025; BizTech, 2025).

Data Characteristics and Segmentation Potential

Key variables include:

- **Inquiry frequency:** Number of times a topic appears across support channels
- **Inquiry complexity:** Proxy measures such as average word count, escalation rate, or resolution time
- **Topic type:** Categorical labels (e.g., insurance, app connectivity, billing)
- **Channel source:** Chat, phone, or web

These variables support segmentation and pattern recognition across ADC’s support content.

Feature extraction is critical. To apply cluster analysis, ADC would need access to: raw text from FAQs, chat logs, and call transcripts, metadata such as timestamp, channel, and frequency, and search query logs from the website and app. As Yeomans notes, clustering often begins with identifying co-occurring terms and topics. For ADC, this could mean grouping inquiries about sensor adhesion, insurance coverage, or app connectivity. These features can be derived from natural language processing techniques such as tokenization, TF-IDF weighting, and semantic embedding.

Ranked Modeling Approaches (Chan, 2025; Appendices A, B, C, D):

Model	Strengths	Limitations	Rank
Decision Trees	Intuitive, handles categorical and continuous data, easy to interpret	Risk of overfitting without pruning	1
Support Vector Machines (SVM)	High accuracy, good for nonlinear separation	Less transparent, harder to interpret for business stakeholders	2
Logistic Regression	Simple, statistically grounded, good for binary classification	Limited with complex or nonlinear data	3

Random Forests	Robust ensemble method, reduces overfitting, handles mixed data	Less interpretable, more resource-intensive	4
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Final Recommendation: Decision Trees are the most suitable model for ADC’s support inquiry classification. They balance interpretability, flexibility, and performance—making them ideal for operational deployment and stakeholder trust. Unlike more opaque models, decision trees allow business and regulatory teams to trace how classifications are made, which is critical in a healthcare environment governed by HIPAA and FDA standards. With proper pruning and validation, they can guide content updates, agent training, and escalation protocols. This aligns with recent findings that decision trees and clustering are increasingly used to optimize support workflows and reduce content sprawl (IEEE Xplore, 2025).

Strategic Impact: Cluster analysis supports ADC’s strategic goals in several ways:

- **Improved user experience:** Smarter search and navigation reduce friction for users managing chronic conditions.
- **Operational efficiency:** Identifying redundant content lowers copywriting and MLR review costs.
- **AI-assisted support:** Clusters can inform agent-assist tools and proactive guidance systems.
- **Scalability:** As ADC expands access to CGM systems, clustered insights help tailor support for diverse user segments.

This reflects Chan’s (2025) view that analytics should be used to “match internal strengths to external pressures.” ADC’s regulatory expertise and commitment to Caring make it well-positioned to deploy unsupervised learning responsibly.

Risks and Considerations: While unsupervised learning offers flexibility, it also requires careful interpretation. Clusters may reflect noise or bias if not validated. Regularization techniques can help reduce overfitting (Yeomans, 2015). Cross-validation - testing clusters on withheld data - ensures generalizability and guards against misleading patterns. Additionally, ADC must ensure HIPAA compliance and maintain transparency in how support content is grouped and presented. As Davenport and Harris note, analytics must be “renewable, adaptable, and embedded” - not just technically sound.

- **Ensuring Value:** The design should specify the quality threshold of the end nodes in the decision tree to ensure usefulness and value (ie, minimum number of items in each node, greater than 2 or less than 10, for example).
- **Relevance & Currency:** this should be an ongoing process of continuous training to incorporate user feedback and updated information per MLR and product needs.
- **Modularity & Scalability:** Design should consider future integrations for flexibility and scalability. For example, connecting the Veeva Promomats system used to manage content for MLR and other “output” systems such as labelling and the web content management system.

Cross-Functional Collaboration: Implementing cluster analysis requires coordination across multiple departments including IT, Customer Support, Marketing, Regulatory/MLR and UX. ADC should consider a dedicated change management analyst to work in parallel with the program manager to ensure all impacted systems and users are integrated and aligned for this project. This approach aligns with Davenport and Harris’s (2007) emphasis on embedding analytics into core business processes. ADC’s ability to act on clustered insights depends on organizational capability - not just technical accuracy. Implementation of data analytics should be iterative and included from the initial design to ensure that efficiency and outcomes of the decision tree cluster model can be tracked and adjusted.

Conclusion: Cluster analysis offers a strategic, low-risk way for ADC to improve its customer support capability during the new design system migration. By organizing unlabeled support content into meaningful clusters, ADC can enhance user experience, reduce internal costs, and align with its values of safety, trust, and care. This is a clear example of how unsupervised machine learning can transform a business capability - not through prediction, but through understanding.

APPENDIX A: Sample Decision Trees

“Decision tree or recursive partitioning is a supervised graph based algorithm to represent choices and the results of the choices in the form of a tree.”

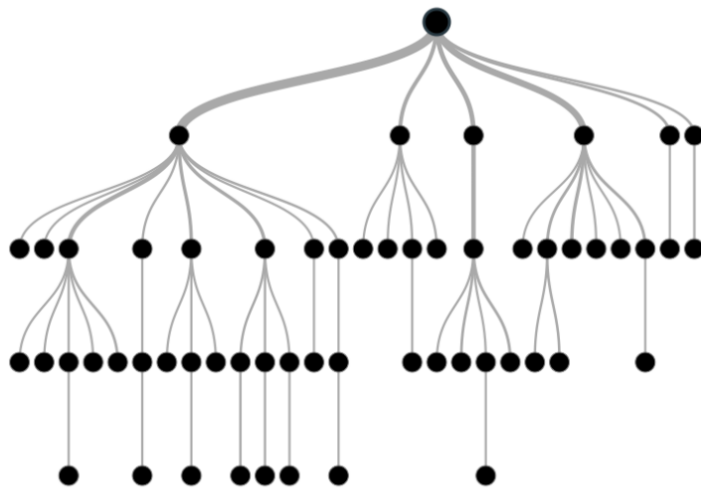


Figure 7.1: Decision Tree

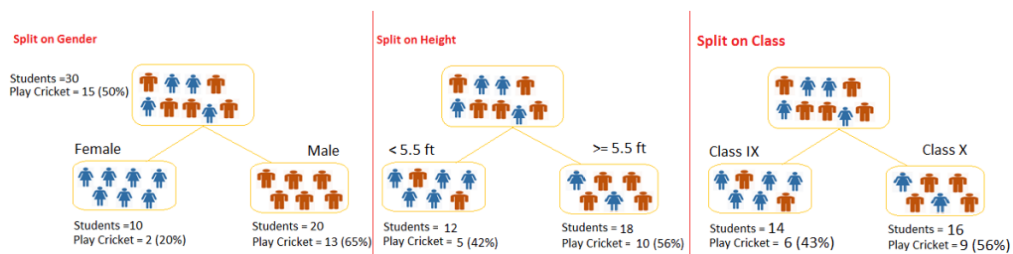


Figure 7.2: Example of a decision Tree

IMAGE SOURCE: <https://cambiotraining.github.io/intro-machine-learning/decision-trees.html#decision-trees-1>

APPENDIX B: Support Vector Machines

Support Vector Machine (SVM)

Support Vector Machine or SVM is used for classification. It finds the best boundary or hyperplane that separates data into different classes. The goal is to maximize the margin between the closest data points of each class. Using an analogy, if you and your friends have different types of candies, SVM is like drawing a line that sorts out all the candies into their respective groups.

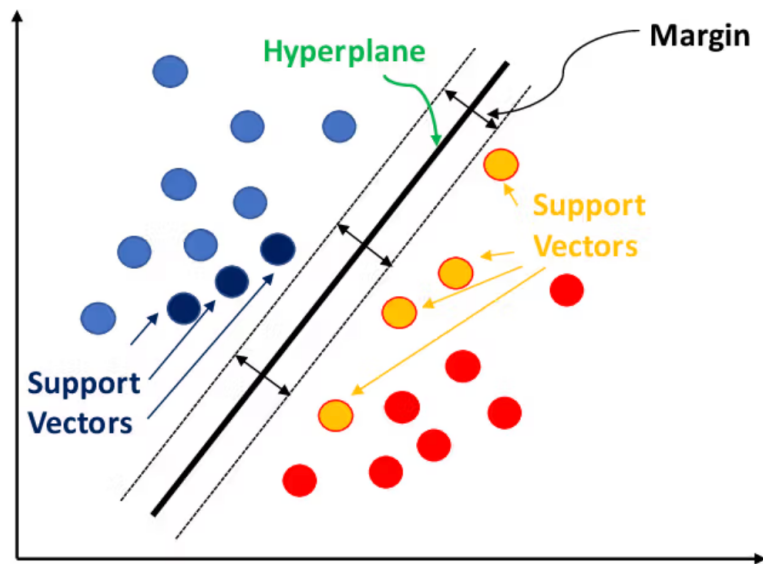
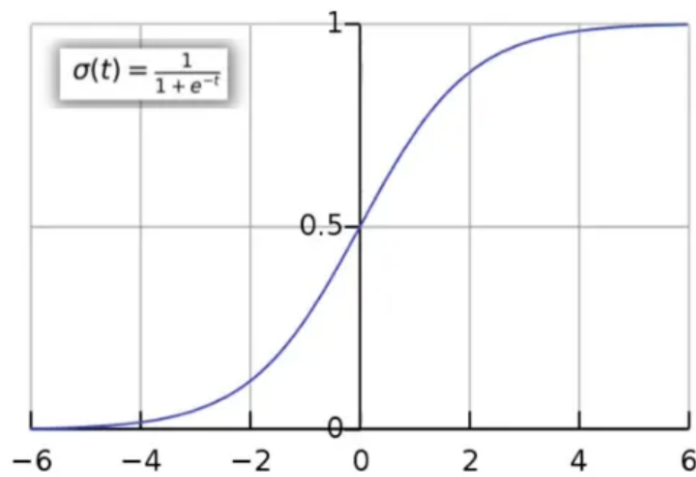


IMAGE SOURCE: https://www.stratascratch.com/blog/a-comprehensive-overview-of-3-popular-machine-learning-models/?#support_vector_machine_svm

APPENDIX C: LOGISTIC REGRESSION



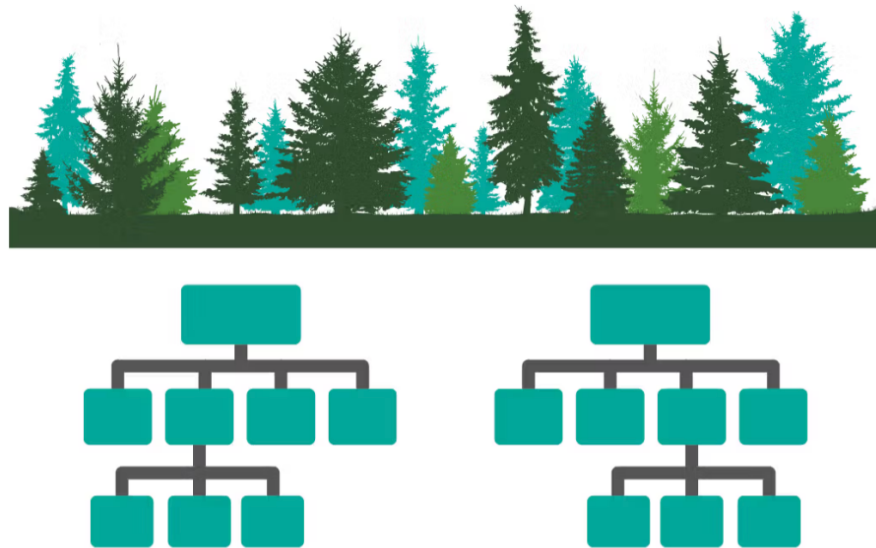
Logistic regression with sigmoid function (image by author).

Logistic regression can be used in several classification cases, such as binary classification (pass or fail), multiclass classification (cat, dog, or pig), and ordinal classification (high, medium, low). To evaluate the logistic regression model, we can use the confusion metric, ROC Curve, AIC (Akaike Information Criteria), Null Deviance, and Residual Deviance.

IMAGE SOURCE: <https://dedekurniawann.medium.com/popular-machine-learning-algorithms-supervised-and-unsupervised-learning-766120e96d49>

APPENDIX D: Random Forests

Random forest is probably one of the most popular decision tree models. It is an ensemble of decision trees. It creates different sets of decision trees from randomly selected data points and then averages their predictions. This reduces the risk of overfitting and improves overall accuracy.



SOURCE: <https://www.stratascratch.com/blog/a-comprehensive-overview-of-3-popular-machine-learning-models/>

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